

ASSIGNMENT INSTRUCTIONS: PBIT 206

Subject: Advanced Data Communication

Assignment Type: Research Essay

Status: Mandatory

1. GENERAL SPECIFICATIONS

- **Title:** Advanced Data Communication: Principles and Applications of Satellite
 - **Word Count:** 8,000 words (minimum)
 - **Font Type:** Times New Roman
 - **Font Size:** 12 pt
 - **Line Spacing:** 1.5
 - **Alignment:** Justified
 - **Referencing Style:** APA 7th Edition
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2. TASK DESCRIPTION

The student is required to produce a comprehensive, well-researched essay examining the architectural and functional complexities of satellite data communication. The work must demonstrate an expert-level understanding of how satellite systems facilitate global data exchange, with a specific focus on "Satellite Zooming" (High-Throughput Spot Beam technology) and high-resolution data acquisition.

3. REQUIRED CONTENT COVERAGE

To meet the 8,000-word requirement, the essay must be subdivided into the following critical areas:

- **Introduction:** Evolution of satellite communication and its role in the modern global data infrastructure.
- **Technical Principles:** Analysis of orbital mechanics (LEO, MEO, GEO), frequency spectrum management (C, Ku, Ka bands), and signal modulation.
- **Satellite Zooming & Precision:** Technical exploration of High-Throughput Satellites (HTS), frequency reuse, and targeted spot beam delivery.
- **Comparative Analysis:** Advantages and disadvantages of satellite vs. terrestrial data communication.

- **Challenges:** Detailed discussion on atmospheric attenuation, latency, orbital debris, and cybersecurity.
- **Future Trends:** Integration with 6G networks and AI-driven satellite autonomous operations.

4. RESEARCH & CITATION REQUIREMENTS

- **Minimum Citations:** 0 properly formatted sources.
- **Source Quality:** Only academic journals, conference papers (e.g., IEEE), and official industry white papers are permitted.
- **Compliance:** In-text citations and the final reference list must strictly follow **APA 7th Edition** guidelines.

5. SUBMISSION PROTOCOL

URGENT: All students must ensure their **Turnitin** accounts are activated immediately. This essay must be submitted via Turnitin; **no other platforms or manual submissions will be accepted.** Failure to activate your account in time will not be considered a valid excuse for late submission.

6. GRADING SCALE & RUBRIC (Total: 40 Marks)

The essay will be evaluated based on the following criteria. To achieve a high score, students must demonstrate a mastery of data communication principles and strictly follow the 8,000-word mandate.

Category	Criteria	Marks
Technical Depth & Accuracy	Mastery of satellite data communication principles, link budgets, and orbital mechanics. Detailed analysis of "Satellite Zooming" and spot beam technology.	15
Research & APA Compliance	Successful integration of 50+ peer-reviewed sources . Precise application of APA 7th Edition for both in-text citations and the reference list.	10
Word Count & Scope	Adherence to the 8,000-word requirement. Breadth of coverage across all required chapters without "filler" content.	8
Organization & Structure	Logical progression of ideas, effective use of headings, and a professional academic tone. Clear introduction and comprehensive conclusion.	4
Formatting & Presentation	Strict adherence to Times New Roman (12pt), 1.5 Spacing , and Justified alignment . No grammatical or typographical errors.	3

Category	Criteria	Marks
TOTAL		40

Grade Classification:

- **32 – 40 Marks (First Class / Distinction):** Demonstrates expert-level knowledge. All 50+ citations are relevant and correctly formatted. Word count is met with high-quality technical discussion.
- **24 – 31 Marks (Upper Second Class):** Strong grasp of the topic. Meets the citation and word count requirements but may have minor technical gaps or formatting inconsistencies.
- **20 – 23 Marks (Pass):** Minimum requirements met. Basic understanding of satellite systems shown, but the 8,000-word expansion may lack sufficient depth or analytical rigor.
- **0 – 19 Marks (Fail):** Submission fails to meet the 8,000-word threshold, lacks the minimum 50 citations, or shows poor technical understanding.

Final Submission Warning:

Turnitin account activation is a prerequisite for submission. Any assignment not processed through the official Turnitin portal will be awarded **0 marks**. Ensure your login credentials are functioning correctly well in advance of the deadline.